```
"""Main execution script for RIPS-frontier demonstration."""

from rips_frontier import RipsFrontier


def main():
    print("=== Initializing RIPS-Frontier ===")
    # Initialize the graph with source node 's' and a vertex capacity
of 10
    rf = RipsFrontier(s="s", n_vertices=10)

    print("\n--- Step 1: Inserting Edges (Building a path: s -> a -> b
-> t) ---")
    rf.insert("s", "a")
    rf.insert("a", "b")
    rf.insert("b", "t")

    # Run queries
    print("Is 't' reachable from 's'?", rf.query("t"))  # Expected:
True
    print("Is 'x' reachable from 's'?", rf.query("x"))  # Expected:
False

    print("\n--- Step 2: Adding an alternative path (s -> c -> t)
---")
    rf.insert("s", "c")
    rf.insert("c", "t")
    print("Is 't' reachable now?", rf.query("t"))  # Expected: True

    print("\n--- Step 3: Deleting a tree arc (s -> a) to test
pruning/repair ---")
    rf.delete("s", "a")

    # Check if 'b' is still reachable
    print("Is 'b' still reachable after deleting s->a?",
rf.query("b"))  # Expected: False

    # Check if 't' is still reachable via the alternative path (s -> c
-> t)
    print("Is 't' still reachable via alternative path?",
rf.query("t"))  # Expected: True

    print("\n=== Execution Summary ===")
    print(f"Total operations processed: {rf.ops}")
    print(f"Current operating mode: {rf.mode}")
    print(f"Total abstract work units: {rf.total_work}")
```

```
if __name__ == "__main__":
    main()
```